

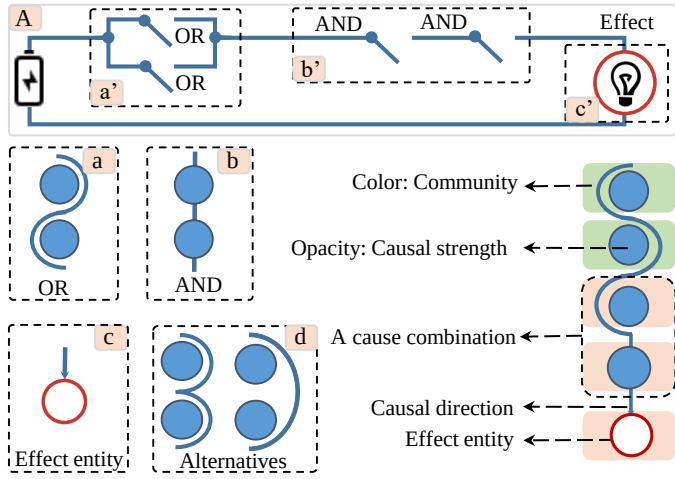


Fig. 5. Visual metaphors of “electrocircuit” (A) for illustrating the aggregated causality relationships. An arc surrounding a circle serves as an optional causal element (“OR”), a line crossing a circle encodes the combined causal element (“AND”), and a red circle encodes a effect entity.

Visual metaphor (T4). Designing a formative and intuitive visual metaphor for multiple combined causes is an effective and informative method for users to obtain the overall distribution of causal relations. As illustrated in Figure 5(A), multiple causality are aggregated as an informative visual metaphor of “electrocircuit”, which is a significant visual component of our visual causal analytics system. Operations of merging optional causes (“OR”) and indispensable causes like functions of switches in a parallel circuit and a series circuit. A closed circuit can light a light bulb. Causes in Figure 5(a) and (b) represent “AND” or “OR” switches in Figure 5(a’) and (b’), respectively. An effect entity in the lower left corner of Figure 5(c) represent a light in Figure 5(c’). An arc surrounding a circle serves as an optional causal element (“OR”), a line crossing a circle encodes the combined causal element (“AND”), and a red circle encodes a effect entity. The right corner of the Figure 5 illustrates three pairs of causality, where a line crossing a circle is a indispensable cause element (“AND”) and an arc surrounding three circles are optional causes element (OR). A “AND” circle combining with a “OR” circle is a cause combination for a effect entity, such as a cause combination in the black dotted area. In addition, two visual metaphors (d) in the lower left corner of Figure 5 are our alternative designs, which tend to lack of pleasing and lead cognition bias on a cause combination. Thus, we utilize a pleasing and smoothing curve around optional causes, as well as a direct line crossing indispensable causes.

Nodes binding and aggregating (T4, T5). Filtering and aggregating vertices to streamline graphs are common approaches to ensure readability and informativeness of causal hypergraphs, such as merging hyperedges that share the same entity. In our work, cause combinations are aggregated as visual metaphors of “electrocircuit” to optimize the horizontal space, eliminate demanding context switching and ensure scalability. As illustrated in Figure 6, the matrix graph is the original causal hypergraph, where horizontal circles describe the same entity aligned on the left side, the solid and hollow circles encode cause entities and effect entities, respectively. Solid green or purple circles serve as the combined causes and red hollow circle represents the effect entity. A line with arrow connecting cause entity and effect entity means a pair of causal relationship. The visual metaphor of “electrocircuit” in the middle of the Figure 6(a’)(b’) are the aggregation results of original causal hypergraph(a)(b). We utilize an arc surrounding circles to indicate optional causes (“OR”) in combination and a line crossing circles to encode indispensable causes (“AND”) in combination. Figure 6(a) illustrates four causes, which can independently affect an effect entity respectively. To save the horizontal space of edges, four pairs of causal relations are packed and collapsed in figure 6(a’), which saves the horizontal space of three edges. In the same way, Figure 6(b) displays four pairs of causality. Entity *F* can be combined with the entity *G*, *H*,

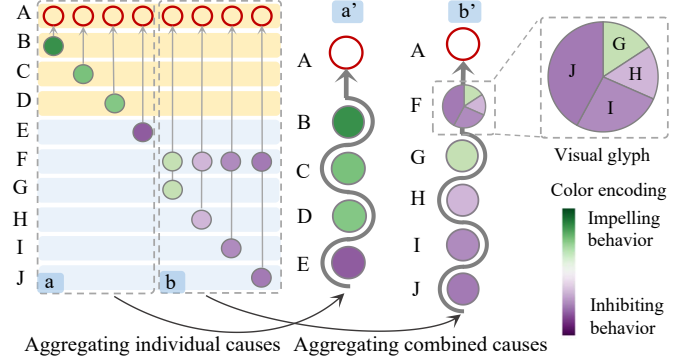


Fig. 6. Principles of causality aggregation to ensure informativeness and compact of similar cause combinations. Left original causality (a)(b) are aggregated to the right metaphors (a’)(b’).

I, or *J* to affect entity *A*. These four pairs of causal relations are aggregated in Figure 6(b’). In addition, Pie chart in the upper right corner of Figure 6(a) is utilized to illustrate the characteristic of aggregated causality. The radian size of each sector represents causal strength and the color of each sector represent impelling or inhibiting behaviors.

Multi-level exploration (T6). Depicting causality at a different level of causal abstraction not only provides the distribution of causal information, but also the details of each pair of causality. Figure 1 (A) shows that the focused area (a) provides and enhances a user’s concerned facts. Contextual area (c) is always necessary to enhance the expressiveness of a visualization. Transition area (b) provides a smoothing effect for aggregation/disaggregation from the context area to the focused one. Controller panel (d) is designed in spired by Multistream [9], including brushing and linking techniques to help users handle and expand focus or context area. The information in a gray area in the controller is zoomed and expanded in the focused area. The causal relations in the focused area are expanded by dragging the controller. When two sides of the controller are dragged, the gray area in the controller is broadened. When the focused area in the controller is dragged, the length of the contextual area is dynamically updated. The focused area is broadened by dragging the two sides of the controller.

5.2 Auxiliary Visualization for Causality Exploration

Confirming and explaining impelling and inhibiting behaviors of causality are necessary, such as the navigation of amending the incorrect causal relations and the influence propagation of the entities.

Causality propagation visualization. Users always need to drill down into the causality and explore the influence propagation in large searching space. Causality propagation reveals how the stimulus propagates over and influences entities. The roots in a causal graph imply sources and the pathways uncover how a source entity propagates over and influences entities. Causality queries and propagation visualizations are supported in our system by clicking visual encoding of cause and effect entities. We employ Dijkstra algorithm [33] and Stratisfimal layout [12] for identifying and visualizing the shortest influence pathways given the cause and effect entities.

Causality reconfirmation visualization. Considering data noise, uncertainty, and complex interaction among entities, our data-driven causal discovery is not always faithful. The effect entities may be chronologically influenced by a pair of cause entities. We utilize traditional and easy-understanding multiple histograms to visualize the traffic volume information of each entity. To some extend, these multiple histograms can help users confirm certain impelling and inhibiting behaviors of causality and navigate users modifying incorrect causal relations by comparing the chronology of entity occurrence.

5.3 User Interactions

Our work combines various basic and advanced interaction techniques, including multiple views linking, dragging, and brushing, to support and enhance the combined causality analysis.

Exploring causality at a multiple levels (T6). Selecting and handling selected causality segments at different levels of abstraction. All